

LeviCircuits: Adhoc Electrical Circuit Prototyping using Ultrasound Levitation

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CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms**;

KEYWORDS

circuits, rapid prototyping, tele-manipulation, levitation

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1 INTRODUCTION

Many researchers are interested in creating re-configurable circuits where users can create temporary and manipulable electrical connections for applications such as rapid prototyping, debugging, and learning. Traditionally, prototyping with circuit connections is achieved through various direct methods like breadboards, where jumper wires are used to connect or disconnect electrical components. Researchers have further built on these direct prototyping means by supplementing them with digital functionalities and toolkits like Virtual Component [4], ToastBoard [2], and the Visible Breadboard [6], which recognize circuit components and their values to create software-based circuit counterparts.

Re-configurable circuits: More recently, researchers have investigated developing hardware solutions for re-configurable circuits. For example, Stretchable and Soft Electronics [1] uses liquid metal to create shape-configurable circuits through guiding electrically actuated gallium-based liquids on pre-fabricated paths. Similarly, Flowcuits [10] uses blue-tac as a re-usable material to engrave guiding circuit paths for liquid metals to flow on. These direct means, however, require physical intervention, which is sub-optimal for remote learning experiences and collaborations. In addition, direct contact can be dangerous for high risk situations, such as high current circuits or young learners being as users. While researchers like Twafik et. al [9] developed tele-manipulated circuits via controlling the state of switches on a relay matrix, this technique requires desired circuit connections to be physically implemented on the circuit board prior to manipulation.

Levitation of re-configurable components: Finally, several researchers have used acoustic levitation to levitate and manipulate small polystyrene beads for various purposes, including mid-air assemblies of non-electronic components [3] or projection displays [5]. Additionally, some researchers have been able to levitate electronic components; notably, Oichai et. al [7] were able to levitate an LED and a resistor, and Luciola [8] was able to levitate a wirelessly powered circuit chip.

We propose LeviCircuits as a way to mitigate the shortcomings of current circuit prototyping platforms by using ultrasound levitation. LeviCircuits enables:

- (1) Tele-manipulation of electrical circuits in a 3D space for remote learning, debugging, and collaboration through temporarily connecting levitating electrically-conductive nodes.
- (2) The creation of 3-Dimensional circuit connections, in comparison to traditional 2D breadboarding.

2 LEVICIRCUITS

We propose ultrasound levitation as a means to create a playground for re-configurable, tele-manipulated circuits. To do so, we utilize the conductive 3D printing filament, Electrifi, which has a conductivity of 0.06 Ohm-cm, to create conductive spheres which we refer to as "nodes". These nodes, which are attached to very thin copper wires, levitate via ultrasound to connect to other floating nodes. These nodes can also house small electrical components, such as LEDs and resistors (Figure 1). In terms of the interaction pipeline, we envision the user to add or remove nodes from the levitation system. Then, acoustic levitation moves the nodes to specific locations to create electrical connections. Such manipulation of the connections can be done remotely through software, and does not require direct touch. For example, an instructor can aid a student in debugging their circuit remotely by moving their system's nodes, without physically touching the nodes.

3 PRELIMINARY EXPLORATIONS

Next, we present the results from our preliminary tests to evaluate:

- (1) The print quality of the 3D printed nodes and their smallest printable sizes, in comparison to polystyrene beads
- (2) The quality of electrical contact, connectivity, and conductivity assessed by direct electrical contact of a LeviCircuit node pair

Quality of print and smallest print size: We first explored using a Prusa 3D printer with a 0.4mm nozzle diameter to print the conductive spheres. The conductive filament, Electrifi, was purchased from Multi-3D and has a reported conductivity of 0.06 ohm-cm. We conducted a series of smallest print size explorations (diameter:

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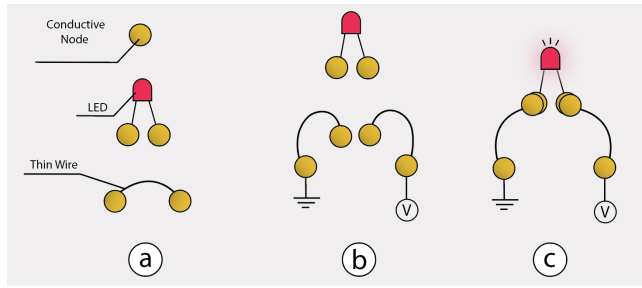


Figure 1: LeviCircuits creates temporary electrical connections via ultrasound levitation: a) the conductive spherical elements, called nodes, attach to thin wires and house electrical components like LEDs. b) When the nodes are placed inside an ultrasound array, they are reconfigured to disconnect or connect from other components. c) Contact between the nodes creates temporary electrical connections between different components

10mm, 7mm, 5mm, 3mm) with 100% infill for best conductivity. The smallest size we successfully printed for the conductive nodes was 7mm in diameter, but we were able to decrease the diameter down to 3mm via post processing methods such as re-heating and manually molding the printed nodes (Figure 2), which allows for wires or other things to be embedded into the nodes as well. In comparison to the polystyrene beads, which have a diameter range of 2mm–3mm, we expect our conductive nodes, with respect to diameter, to be within reasonable range of levitation standards. The net weight of a 3mm bead of Electrifi filament is roughly 0.04 g. Assuming a density of 0.96 g/cm^3 for the polystyrene beads, a 3mm bead is expected to be 0.013g. Our conductive nodes (without any magnets or electronic components) are approximately 3 times heavier than the polystyrene beads. We have also explored coating polystyrene beads with highly conductive grease/paint to enable levitation and conductivity. We discuss these explorations in subsequent sections.

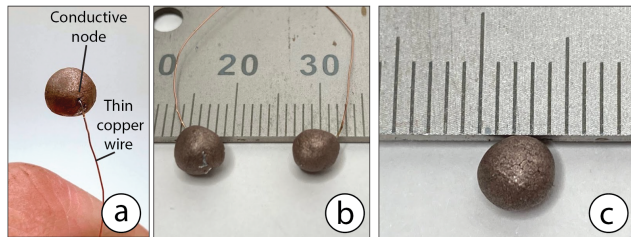


Figure 2: LeviCircuit nodes attached to thin copper wires and are 3–4mm in diameter

Quality of contact: To assess the quality of contact between a pair of conductive nodes, we attached two nodes to a single LED and resistor combination: the first node connected to ground and the resistor, and the second connecting to the LED only (i.e. floating). We then brought the floating node into contact with another floating node that is attached to power (Figure 3). Direct touch

contact between the conductive nodes did not light the LED, which is expected due to the poor quality of contact and high contact resistance of the conductive filament (Figure 3-b). However, when covering the conductive nodes with copper tape, we were able to increase the quality of contact, and subsequently lit the LED. (Figure 3-a)

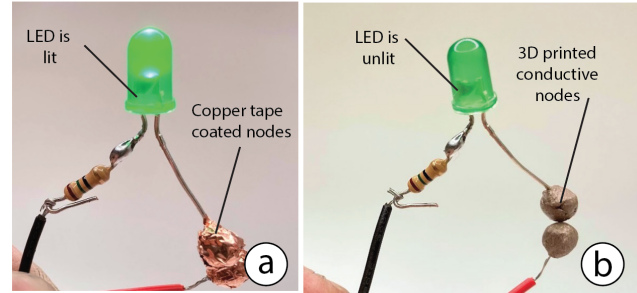


Figure 3: LeviCircuit nodes have better electric contact when coated with copper tape (a) compared to direct contact with Electrifi 3D printing filament without the tape (b).

Quality of connectivity: Even with copper tape, the nodes were unable to sustain the connection for long, especially when we introduced small jitters that we expect to be produced from levitation. To solve this, we embedded weak magnets within our nodes as a means to sustain connections against vibrations. We used 2mm chunks of 675 Gauss magnets. We manually warmed and molded the conductive 3D printing filament around the magnet to achieve this. The magnets successfully sustained contact even when lightly shook.

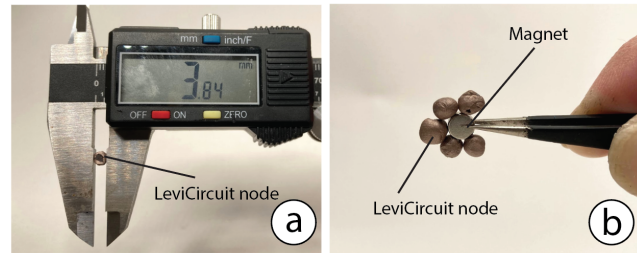


Figure 4: LeviCircuit nodes with magnets: a) Cross section of the magnetic core (black) encased in conductive Electrifi filament. b) A group of magnetic nodes attracted to a diametrically opposed magnet.

Quality of conductivity: Despite the effectiveness of copper tape's conductivity for node connections, perfectly wrapping a planar piece of copper tape around a spherical node is difficult to reproduce consistently. Additionally, the extra weight and volume copper tape adds to the nodes is undesirable for levitation. In effort to explore alternative paths, we investigated the effects of: 1) lithium-based conductive grease with copper particles, 2) copper-based conductive paint, 3) silver-based conductive paint, 4) and silver-based conductive ink for inkjet printing, on the quality of node to

node surface conductivity. Our experiments showed a detectable improvement in conductivity of the LeviCircuit nodes using copper paint (LumiLor BP-311 backplane). Figure 5 shows how the copper-based coat enables the Electrifi-based LeviCircuit nodes to conduct well enough to light an LED. The silver based materials, which required heating after application to make conductive, melted the 3D printing conductive element Electrifi. We have also attempted to coat the polystyrene beads with the conductive inks and paints via 1) spray painting 2) dipping in paint/ink. Both methods did not work as spray painting blew the light beads away, and dipping caused the polystyrene beads to dissolve upon contact.

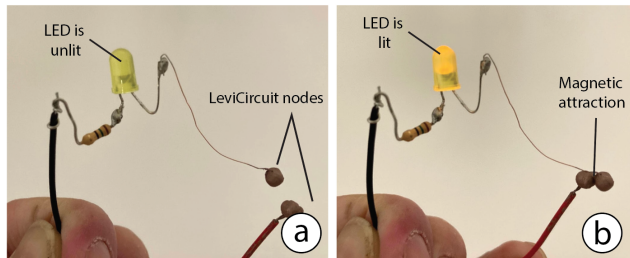


Figure 5: LeviCircuit nodes with magnetic cores and a coat of copper-based conductive paint are able to connect and complete a circuit.

4 NEXT STEPS

First, we plan on finding ways to reduce the size and weight of the nodes to improve levitation, since our nodes with magnetic cores are currently weighing around 0.1g each and are difficult to consistently hand-make smaller than 3mm in diameter. We plan to explore other means of conductive coating and node-to-node connectivity in order to be able to use lighter materials like polystyrene.

Additionally, we plan to explore means of incorporating smaller and lighter electronic components. Rather than using breadboard LEDs and resistors like we have used for testing so far, we plan to use much smaller SMD components that can easily be mounted directly on the nodes (Figure 6).

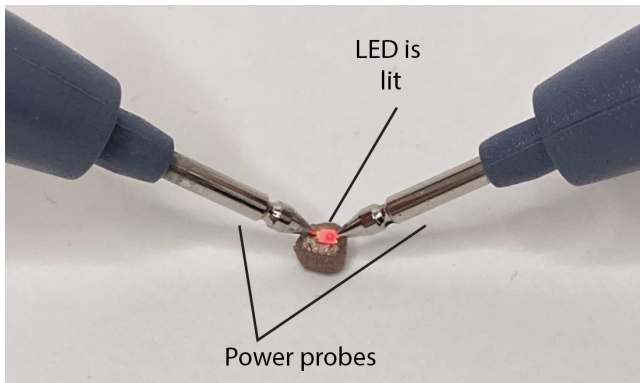


Figure 6: An example of an SMD LED mounted directly on a LeviCircuit node.

Finally, once the quality of contact and levitation are confirmed, we hope to test our prototyping tool with the ultrasound matrix for controlled levitation.

5 CONCLUSION

We propose LeviCircuits as a means of using acoustic levitation to create re-configurable electrical connections. This would enable remote manipulation and prototyping of circuits for either learning or collaboration purposes. Our preliminary explorations demonstrate that it is feasible to create nodes that can maintain connections with one another via magnetic attraction, conduct electricity upon contact with other nodes, and are within the size and mass ranges required to achieve levitation. We hope to use the ultrasound matrix to further test our prototype of levitating re-configurable circuits.

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